

TRI-NETRA ADCS: design loop, SILS, Monte Carlo and soft OILS

From each customer case to a sized, selected and dispatched attitude-control configuration, verified on the Rust engine (POP v51 in the loop), against the MATLAB twin, and with the flight software running as Cortex-M4F firmware.

AIS_3U SELECTED mtq_fmr closest (not feasible)	AIS_IMG_3U SELECTED mtq_fmr closest (not feasible)	ENGINE VS MATLAB VERDICTS 94/107 truth environment identical
SILS VS SOFT OILS VERDICTS 104/107 worst OBC load 6.8 %	VIRTUAL OBC BIT-IDENTICAL 35 runs, process + QEMU	CAMPAIGNS ON THE ENGINE 8 MC + edge, vs MATLAB

Owner	Agastya
Generated	2026-09-28 17:24 UTC
Repository revision	d649535
Engine	adcs-engine-rs/1.2.0 (adcs-case/1, POP v51 port in-loop, adcs-design, soft OILS)
Flight software	trinetra-fsw-c/1.0.0 and trinetra-fsw-rs (C99 and Rust no_std, adcs-fswcfg/1)
Cases	ais_3u, ais_img_3u
Regenerate	<code>python3 tools/pipeline.py · python3 tools/engine.py campaign · python3 tools/engine.py oils · python3 tools/vv_report.py</code>

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1. Summary and findings

ais_3u (coarse class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), closest (not feasible), after 4 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 0.855 kg, 0.477 L. Gaps: budget: `mass_kg` 0.855 > 0.35; budget: `volume_L` 0.477 > 0.3

ais_img_3u (fine class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), closest (not feasible), after 5 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.559 kg, 0.474 L. Gaps: `nadir_pointing: ape_los_p9973` (performance), `rate_stability_p9973` (performance); budget: `mass_kg` 1.56 > 1

2. Architecture and flow

The design loop sizes every actuator option to the case, flies every mission mode with every option in the SILS, converges the sizing, sensors and algorithms, selects the simplest of our solution families that passes, dispatches its flight configuration and verifies it by Monte Carlo and in soft OILS. The same flight software (C99 and Rust, one pseudocode contract) runs in every rung behind one byte-level HAL.

TRI-NETRA ADCS: from customer case to flight

End-to-end V&V flow: design loop, software verification, then hardware-in-the-loop rungs

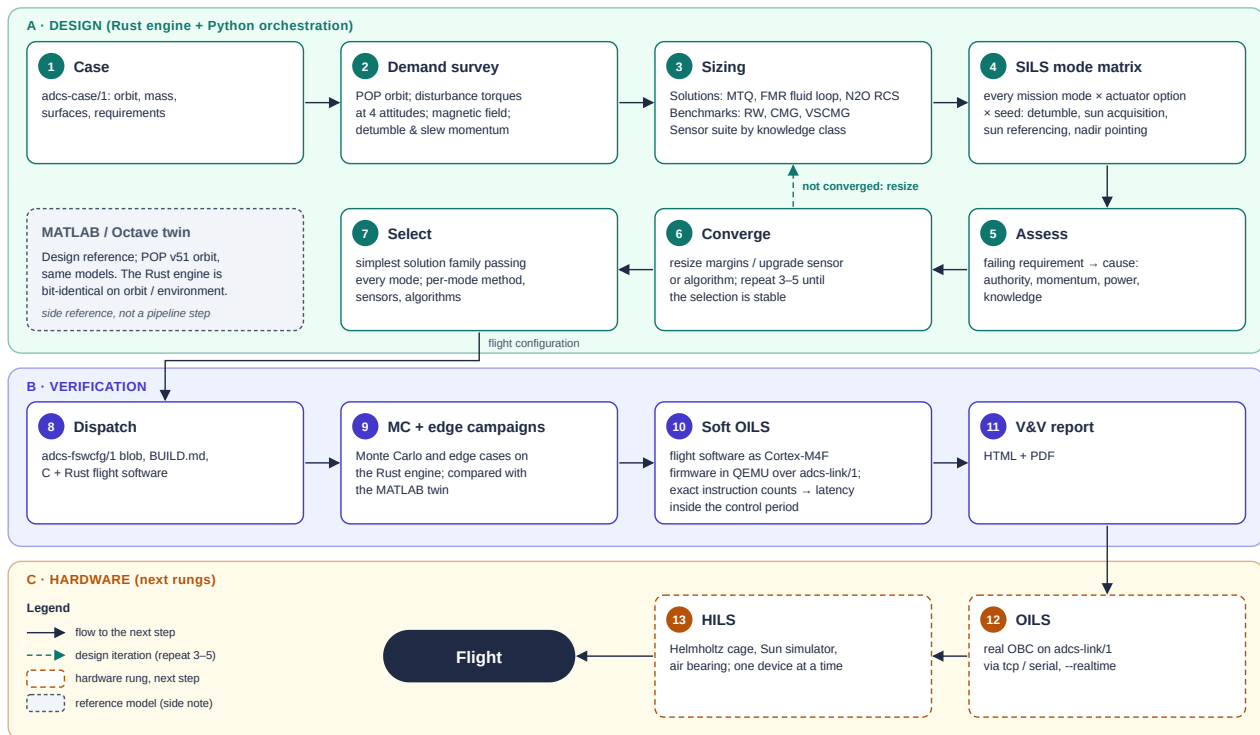
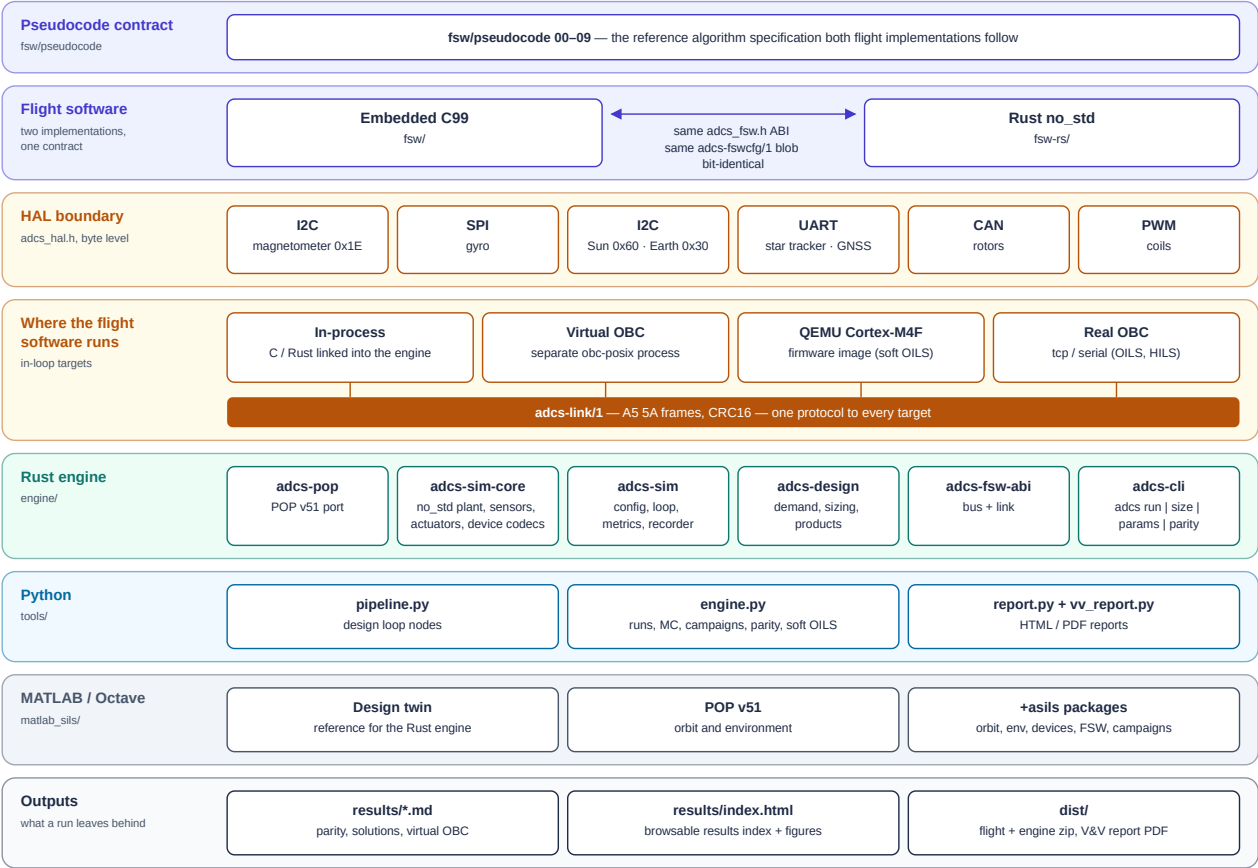


Figure 1. From customer case to flight: design loop, verification, hardware rungs.

Who runs what: languages and layers

TRI-NETRA ADCS, from the flight contract at the top to the ground tooling and outputs at the bottom



TRI-NETRAADCS - Agastya

Figure 2. Languages and layers.

3. Requirements

ais_3u — 3U AIS satellite — coarse pointing 10 deg, SSO dawn-dusk 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	10.0	Degree	stated by the user: the AIS antenna needs only coarse nadir pointing
req.ake	Absolute knowledge error (AKE)	5.0	Degree	UNCONFIRMED · plan reference: half the APE, so knowledge never dominates the budget
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	0.35	Kilogram	UNCONFIRMED · plan reference: a coils-and-magnetometer ADCS in a 3U
req.pavg	ADCS orbit-average power	0.5	Watt	UNCONFIRMED · plan reference: leaves the AIS receiver its share of a 3U power budget
req.ppk	ADCS peak power	1.5	Watt	UNCONFIRMED · plan reference: all coils at full current
req.vol	ADCS volume	0.3	Litre	UNCONFIRMED · plan reference: under a third of 1U
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1
orbit.ltan	LTAN	6.0	Hour	stated by the user: SSO dawn-dusk (LTAN 06:00)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal

key	quantity	value	unit	source / status
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12 \cdot (0.10^2 + 0.10^2)$

ais_img_3u — 3U imaging satellite — fine pointing 0.01 deg, SSO 10:00 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	0.01	Degree	stated by the user; 3-sigma. A flown 3U reference: MinXSS-1 (BCT XACT) measured 0.0042-0.0117 deg 3-sigma, axis dependent (arXiv:1706.06967)
req.ake	Absolute knowledge error (AKE)	0.005	Degree	UNCONFIRMED · plan reference: half the APE
req.rks	Rate stability	0.001	DegreePerSecond	UNCONFIRMED · plan reference: 3.6 arcsec/s, to hold smear under an exposure
req.slew	Reference slew time	60.0	Second	UNCONFIRMED · adcs_ref_c1: mission.sangle in 60 s, target to target
req.settle	Settling time after a slew	20.0	Second	UNCONFIRMED · plan reference: back inside the APE after a slew
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	1.0	Kilogram	UNCONFIRMED · plan reference: wheels, star tracker and coils in a 3U
req.pavg	ADCS orbit-average power	2.0	Watt	UNCONFIRMED · plan reference
req.ppk	ADCS peak power	4.0	Watt	UNCONFIRMED · plan reference
req.vol	ADCS volume	0.6	Litre	UNCONFIRMED · plan reference: about half of 1U
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1

key	quantity	value	unit	source / status
orbit.ltan	LTAN	10.0	Hour	stated by the user: SSO 10:00 (read as LTAN 10:00; LTDN 22:00 gives the same plane and beta magnitude)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$

4. Design loop per case

ais_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (closest (not feasible))

Converged after 4 iteration(s); knowledge class **coarse**; sensors: magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance 5.635e-07 N m (sun), stored momentum requirement 0.001466 N m s, torque requirement 5.757e-05 N m, weakest field 19.76 μ T, detumble momentum 0.00733 N m s.

Open gaps of the selected family: budget: mass_kg 0.855 > 0.35; budget: volume_L 0.477 > 0.3

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	1/25	mtq (closest (not feasible))	nadir_pointing/fmr+mtq: power -> permanent-magnet pump yoke on the fluid loop nadir_pointing/mtq: fly every mtq_pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm sun_referencing/vscmg+mtq: fly every pointing algorithm sun_referencing/vscmg+rscs: fly every pointing algorithm	18
2	fmr_pm	6/25	mtq_fmr (closest (not feasible))	mtqp: authority x1 -> x1.5 (performance)	16
3	mtqp x1.5, fmr_pm	6/25	mtq_fmr (closest (not feasible))	mtqp: authority back to x1 (no improvement)	17
4	mtqp x1, fmr_pm	6/25	mtq_fmr (closest (not feasible))	—	17

Stopped because: detumble/rscs: power fails at the sized authority (rscs); the part's standby power is the floor; mtqp: more authority did not reduce the performance violation (39.9 -> 41); kept at x1; nadir_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor; nadir_pointing/cmg+rscs: power fails at the sized authority (cmg); the part's standby power is the floor; nadir_pointing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor; nadir_pointing/rw+rscs: power fails at the sized authority (rw); the part's standby power is the floor; nadir_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor; nadir_pointing/vscmg+rscs: power fails at the sized authority (vscmg); the part's standby power is the floor; sun_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor; sun_acquisition/rw: power fails at the sized authority (rw); the part's standby power is the floor; sun_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor; sun_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor; sun_referencing/cmg+rscs: power fails at the sized authority (cmg); the part's standby power is the floor; sun_referencing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor; sun_referencing/rw+rscs: power fails at the sized authority (rw); the part's standby power is the floor; sun_referencing/vscmg+mtq: performance and power both fail — no authority change helps; sun_referencing/vscmg+rscs: power fails at the sized authority (vscmg); the part's standby power is the floor

Every family, scored the same way (benchmarks for comparison only):

family	role	feasible	mass [kg]	power [W]	volume [L]	gaps
mtq	solution	FAIL	0.236	2.06	0.0183	sun_acquisition: sun_acquisition_time (performance), sun_angle_p95 (performance); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: ape_los_p9973 (performance)
mtq_fmr	solution	FAIL	0.855	1.88	0.477	budget: mass_kg 0.855 > 0.35; budget: volume_L 0.477 > 0.3
mtq_fmr_rcs	solution	FAIL	1.37	1.93	1.46	budget: mass_kg 1.37 > 0.35; budget: volume_L 1.46 > 0.3
mtq_rw	benchmark	FAIL	0.476	2.56	0.141	sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.476 > 0.35
mtq_rw_rcs	benchmark	FAIL	0.995	2.61	1.12	sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.995 > 0.35; budget: volume_L 1.12 > 0.3
mtq_cmeg	benchmark	FAIL	0.56	2.75	0.275	sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.56 > 0.35
mtq_cmeg_rcs	benchmark	FAIL	1.08	2.8	1.26	sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 1.08 > 0.35; budget: volume_L 1.26 > 0.3
mtq_vscmeg	benchmark	FAIL	0.595	3.15	0.275	sun_acquisition: power_mean (power); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: power_mean (power); budget: mass_kg 0.595 > 0.35
mtq_vscmeg_rcs	benchmark	FAIL	1.11	3.2	1.26	sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 1.11 > 0.35; budget: volume_L 1.26 > 0.3

Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	48.74 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.292 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.2469 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2

mode	option	feasible	objective (worst seed)	algorithms
nadir_pointing	fmr+mtq	pass	0.1478 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, kp, accommodation, reflectivity, arg_lat_deg):

metric	req	mean ± std	range	pass rate
detumble_time	284	56.01 ± 22.5	[9.575, 93.17]	100 %
ape_los_p9973	10	0.2345 ± 0.0759	[0.1372, 0.383]	100 %
ake_los_p9973	5	0.2329 ± 0.0755	[0.1368, 0.3821]	100 %
power_mean	0.5	0.05478 ± 0.00278	[0.04977, 0.05914]	100 %
power_peak	—	0.1756 ± 0.00996	[0.1649, 0.1955]	—
propellant	—	0 ± 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	54.86 pass	54.79 pass	54.83 pass
ape_los_p9973	10	0.1241 pass	0.1232 pass	0.1276 pass
ake_los_p9973	5	0.1237 pass	0.1222 pass	0.1274 pass
power_mean	0.5	0.05488 pass	0.0548 pass	0.0547 pass
power_peak	—	0.1814 —	0.1705 —	0.1735 —
propellant	—	0 —	0 —	0 —

ais_img_3u: selected mtq_fmr — Magnetorquers + fluid momentum loop (closest (not feasible))

Converged after 5 iteration(s); knowledge class **fine**; sensors: star_tracker, magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance 4.367e-07 N m (sun), stored momentum requirement 0.001552 N m s, torque requirement 5.757e-05 N m, weakest field 23.63 μT, detumble momentum 0.00733 N m s.

Open gaps of the selected family: nadir_pointing: ape_los_p9973 (performance), rate_stability_p9973 (performance); budget: mass_kg 1.56 > 1

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	13/25	mtq_fmr (closest (not feasible))	nadir_pointing/cmg+mtq: fly every pointing algorithm nadir_pointing/cmg+rscs: fly every pointing algorithm nadir_pointing/fmr+mtq: fly every pointing algorithm nadir_pointing/fmr+rscs: fly every pointing algorithm nadir_pointing/fmr+rscs: power -> permanent-magnet pump yoke on the fluid loop nadir_pointing/mtq: fly every mtq_pointing algorithm nadir_pointing/rw+mtq: fly every pointing algorithm nadir_pointing/rw+rscs: fly every pointing algorithm nadir_pointing/vscmg+mtq: fly every pointing algorithm nadir_pointing/vscmg+rscs: fly every pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm	2
2	fmr_pm	14/25	mtq_fmr (closest (not feasible))	cmg: authority x1 -> x1.5 (performance) fmr: authority x1 -> x1.5 (performance) mtqp: authority x1 -> x1.5 (performance) rw: authority x1 -> x1.5 (performance) vscmg: authority x1 -> x1.5 (performance)	1
3	cmg ×1.5, fmr ×1.5, mtqp ×1.5, rw ×1.5, vscmg ×1.5, fmr_pm	14/25	mtq_fmr (closest (not feasible))	cmg: authority back to x1 (no improvement) mtqp: authority back to x1 (no improvement) rw: authority back to x1 (no improvement) fmr: authority x1.5 -> x2.25 (performance) vscmg: authority x1.5 -> x2.25 (performance)	4
4	cmg ×1, fmr ×2.25, mtqp ×1, rw ×1, vscmg ×2.25, fmr_pm	14/25	mtq_fmr (closest (not feasible))	fmr: authority back to x1.5 (no improvement) vscmg: authority back to x1.5 (no improvement)	6
5	cmg ×1, fmr ×1.5, mtqp ×1, rw ×1, vscmg ×1.5, fmr_pm	14/25	mtq_fmr (closest (not feasible))	—	6

Stopped because: cmg: more authority did not reduce the performance violation (10.2 -> 10.2); kept at x1; fmr: more authority did not reduce the performance violation (32.1 -> 67.5); kept at x1.5; mtqp: more authority did not reduce the performance violation (1.35e+04 -> 1.35e+04); kept at x1; nadir_pointing/mtq: knowledge fails with the star tracker fitted; rw: more authority did not reduce the performance violation (6.66 -> 6.72); kept at x1; vscmg: more authority did not reduce the performance violation (6.9 -> 6.66); kept at x1.5

Every family, scored the same way (benchmarks for comparison only):

family	role	feasible	mass [kg]	power [W]	volume [L]	gaps
mtq	solution	FAIL	0.83	3.9	0.0118	sun_acquisition: sun_acquisition_time (performance), sun_angle_p95 (performance); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: ape_los_p9973 (performance), ake_los_p9973 (knowledge), rate_stability_p9973 (performance)
mtq_fmr	solution	FAIL	1.56	3.85	0.474	nadir_pointing: ape_los_p9973 (performance), rate_stability_p9973 (performance); budget: mass_kg 1.56 > 1
mtq_fmr_rcs	solution	FAIL	2.02	3.9	1.32	nadir_pointing: ape_los_p9973 (performance), rate_stability_p9973 (performance); budget: mass_kg 2.02 > 1; budget: volume_L 1.32 > 0.6
mtq_rw	benchmark	FAIL	1.08	4.5	0.142	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.08 > 1
mtq_rw_rcs	benchmark	FAIL	1.55	4.55	0.983	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.55 > 1; budget: volume_L 0.983 > 0.6
mtq_cmg	benchmark	FAIL	1.16	4.68	0.272	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.16 > 1
mtq_cmg_rcs	benchmark	FAIL	1.63	4.73	1.11	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.63 > 1; budget: volume_L 1.11 > 0.6
mtq_vscmg	benchmark	FAIL	1.23	5.14	0.316	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.23 > 1
mtq_vscmg_rcs	benchmark	FAIL	1.7	5.19	1.16	nadir_pointing: rate_stability_p9973 (performance); budget: mass_kg 1.7 > 1; budget: volume_L 1.16 > 0.6

Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	63.54 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.275 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.01472 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
nadir_pointing	fmr+mtq	FAIL	0.01211 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, accommodation, reflectivity, arg_lat_deg):

metric	req	mean ± std	range	pass rate
detumble_time	284	73.6 ± 15.5	[44.66, 92.64]	100 %
ape_los_p9973	0.01	0.01188 ± 0.000505	[0.01126, 0.0128]	0 %
ake_los_p9973	0.005	0.00281 ± 0.000332	[0.002367, 0.003611]	100 %
power_mean	2	0.04135 ± 0.00194	[0.03819, 0.04514]	100 %
power_peak	—	0.1211 ± 0.00332	[0.1171, 0.1293]	—
propellant	—	0 ± 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	82.17 pass	82.76 pass	82.74 pass
ape_los_p9973	0.01	0.01247 FAIL	0.01249 FAIL	0.01252 FAIL
ake_los_p9973	0.005	0.003124 pass	0.003113 pass	0.00312 pass
power_mean	2	0.04263 pass	0.04342 pass	0.04343 pass
power_peak	—	0.125 —	0.125 —	0.1255 —
propellant	—	0 —	0 —	0 —

5. SILS scenario verification

Every scenario on the Rust engine with the C flight software behind the byte HAL (POP in the loop). 20 of 40 scenarios meet every requirement they carry; the failures are findings of the design (section 10), not engine errors: the same verdicts come out of the MATLAB twin (section 7).

scenario	what	product	duration [s]	requirement metrics: value / req
agile_slew_cmgy	3U imaging — (cmgy): 90 deg pitch in 15 s	TRN-P-3U-CMG	600	ape_los_on_target_p9973 0.004544 / 0.01 pass
agile_slew_fmr_rcs	3U imaging — (fmr_rcs): 90 deg pitch in 15 s	TRN-P-3U-FMR-RCS	600	ape_los_on_target_p9973 0.006614 / 0.01 pass
agile_slew_img	3U imaging — (img): 90 deg pitch in 15 s	TRN-P-3U-IMG	600	ape_los_on_target_p9973 0.008686 / 0.01 pass
agile_slew_rw_rcs	3U imaging — (rw_rcs): 90 deg pitch in 15 s	TRN-P-3U-RW-RCS	600	ape_los_on_target_p9973 0.008702 / 0.01 pass
agile_slew_vscmg	3U imaging — (vscmg): 90 deg pitch in 15 s	TRN-P-3U-VSCMG	600	ape_los_on_target_p9973 0.01008 / 0.01 FAIL
detumble_ais	3U AIS — B-dot detumble from a 10 deg/s tumble (coils only)	TRN-P-3U-AIS	1.722e+04	detumble_time 37.79 / 284 pass power_mean 0.0217 / 0.5 pass power_peak 0.5502 / 1.5 pass
detumble_ais_bangbang	3U AIS — detumble, B-dot law bangbang	TRN-P-3U-AIS	1.722e+04	detumble_time 32.21 / 284 pass power_mean 0.7114 / 0.5 FAIL power_peak 0.72 / 1.5 pass
detumble_ais_mag	3U AIS — detumble, B-dot law mag	TRN-P-3U-AIS	1.722e+04	detumble_time 41.16 / 284 pass power_mean 0.07839 / 0.5 pass power_peak 0.6377 / 1.5 pass
detumble_img	3U imaging — B-dot detumble from a 10 deg/s tumble	TRN-P-3U-IMG	1.722e+04	detumble_time 47.54 / 284 pass
fault_coil_ais	3U AIS — X coil fails during detumble	TRN-P-3U-AIS	1.722e+04	detumble_time 82.17 / 284 pass power_mean 0.01821 / 0.5 pass power_peak 0.5502 / 1.5 pass
fault_gimbal_cmgy	3U imaging — one CMG gimbal sticks during fine hold	TRN-P-3U-CMG	5740	ape_los_p9973 0.005509 / 0.01 pass ake_los_p9973 0.002624 / 0.005 pass rate_stability_p9973 0.002963 / 0.001 FAIL power_mean 1.61 / 2 pass
fault_gyro_ais	3U AIS — gyro bias step of 0.1 deg/s during nadir hold	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 52.78 / 10 FAIL ake_los_last_orbit 41.92 / 5 FAIL power_mean 0.01056 / 0.5 pass

scenario	what	product	duration [s]	requirement metrics: value / req
fault_st_img	3U imaging — star-tracker head 2 fails during fine hold	TRN-P-3U-IMG	5740	ape_los_p9973 0.1022 / 0.01 FAIL ake_los_p9973 0.1093 / 0.005 FAIL rate_stability_p9973 0.003771 / 0.001 FAIL power_mean 1.557 / 2 pass
fault_wheel_img	3U imaging — reaction wheel 1 fails during fine hold (FDIR + coils)	TRN-P-3U-IMG	5740	ape_los_p9973 28.62 / 0.01 FAIL ake_los_p9973 0.002858 / 0.005 pass rate_stability_p9973 0.6244 / 0.001 FAIL power_mean 1.213 / 2 pass
fine_hold_cmj	3U imaging — fine nadir hold, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	5740	ape_los_p9973 0.005021 / 0.01 pass ake_los_p9973 0.002626 / 0.005 pass rate_stability_p9973 0.002437 / 0.001 FAIL power_mean 1.61 / 2 pass
fine_hold_fmr	3U imaging — fine nadir hold, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.001 FAIL power_mean 0.08406 / 2 pass
fine_hold_fmr_rcs	3U imaging — fine nadir hold, coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.001 FAIL power_mean 0.08406 / 2 pass
fine_hold_img	3U imaging — fine nadir hold, reaction wheels + star tracker (0.01 deg)	TRN-P-3U-IMG	5740	ape_los_p9973 0.01049 / 0.01 FAIL ake_los_p9973 0.002608 / 0.005 pass rate_stability_p9973 0.003552 / 0.001 FAIL power_mean 1.557 / 2 pass
fine_hold_img_lqr	3U imaging — fine nadir hold, LQR controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.004156 / 0.01 pass ake_los_p9973 0.002652 / 0.005 pass rate_stability_p9973 0.006704 / 0.001 FAIL power_mean 1.558 / 2 pass
fine_hold_img_smc	3U imaging — fine nadir hold, SMC controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.01945 / 0.01 FAIL ake_los_p9973 0.002649 / 0.005 pass rate_stability_p9973 0.003482 / 0.001 FAIL power_mean 1.557 / 2 pass

scenario	what	product	duration [s]	requirement metrics: value / req
<code>fine_hold_rw_rcs</code>	3U imaging — fine nadir hold, coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	5740	ape_los_p9973 0.03388 / 0.01 FAIL ake_los_p9973 0.002602 / 0.005 pass rate_stability_p9973 0.005641 / 0.001 FAIL power_mean 1.53 / 2 pass
<code>fine_hold_vscmg</code>	3U imaging — fine nadir hold, coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	5740	ape_los_p9973 0.00828 / 0.01 pass ake_los_p9973 0.002636 / 0.005 pass rate_stability_p9973 0.004762 / 0.001 FAIL power_mean 2.014 / 2 FAIL
<code>mission_ais</code>	3U AIS — integrated chain: detumble then nadir hold (auto mode switch)	TRN-P-3U-AIS	2.87e+04	detumble_time 37.79 / 284 pass ape_los_last_orbit 8.935 / 10 pass ake_los_last_orbit 1.188 / 5 pass
<code>mission_cmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 SGCMG pyramid	TRN-P-3U-CMG	1.722e+04	detumble_time 57.01 / 284 pass ape_los_last_half_orbit_p9973 0.005241 / 0.01 pass
<code>mission_fmr</code>	3U imaging — tumble, detumble, fine hold: coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_fmr_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_img</code>	3U imaging — integrated chain: detumble then fine nadir hold	TRN-P-3U-IMG	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.01066 / 0.01 FAIL
<code>mission_rw_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.007564 / 0.01 pass
<code>mission_vscmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	1.722e+04	detumble_time 57.02 / 284 pass ape_los_last_half_orbit_p9973 0.009595 / 0.01 pass
<code>nadir_hold_ais</code>	3U AIS — nadir hold with magnetorquers only (10 deg requirement)	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 12.64 / 10 FAIL ake_los_last_orbit 5.491 / 5 FAIL power_mean 0.01054 / 0.5 pass
<code>nadir_hold_ais_css</code>	3U AIS — nadir hold with coarse cosine sun sensors (albedo)	TRN-P-3U-AIS-CSS	1.722e+04	ape_los_last_orbit 39.44 / 10 FAIL ake_los_last_orbit 39.22 / 5 FAIL power_mean 0.01064 / 0.5 pass
<code>slew_cmg</code>	3U imaging — 30 deg target slew in 60 s, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	600	settle_time_after_slew 12.4 / 20 pass ape_los_on_target_p9973 0.004233 / 0.01 pass
<code>slew_fmr</code>	3U imaging — 30 deg target slew in 60 s, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass

scenario	what	product	duration [s]	requirement metrics: value / req
slew_fmr_rcs	3U imaging — 30 deg target slew in 60 s, coils + fluid rings + cold-gas RCS	TRN-P- 3U-FMR- RCS	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass
slew_img	3U imaging — 30 deg target slew in 60 s and settle	TRN-P- 3U-IMG	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007925 / 0.01 pass
slew_img_lqr	3U imaging — 30 deg slew, LQR controller	TRN-P- 3U-IMG	600	settle_time_after_slew 0 / 20 pass ape_los_on_target_p9973 0.004439 / 0.01 pass
slew_img_smc	3U imaging — 30 deg slew, SMC controller	TRN-P- 3U-IMG	600	settle_time_after_slew — / 20 FAIL ape_los_on_target_p9973 0.0178 / 0.01 FAIL
slew_rw_rcs	3U imaging — 30 deg target slew in 60 s, coils + reaction wheels + cold-gas RCS	TRN-P- 3U-RW- RCS	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007922 / 0.01 pass
slew_vscmg	3U imaging — 30 deg target slew in 60 s, coils + 4 VSCMG pyramid	TRN-P- 3U- VSCMG	600	settle_time_after_slew 13.2 / 20 pass ape_los_on_target_p9973 0.01083 / 0.01 FAIL
sun_spin_ais	3U AIS — coils-only Sun acquisition: detumble, spin- up, Sun-pointing spin	TRN-P- 3U-AIS	1.722e+04	sun_spin_entry 142.4 / 95 FAIL sun_angle_last_orbit 93.84 / 20 FAIL spin_rate_error_last_orbit 1.116 / 0.5 FAIL sun_spin_share_last_orbit 13.75 / 90 FAIL power_mean 0.45 / 0.5 pass

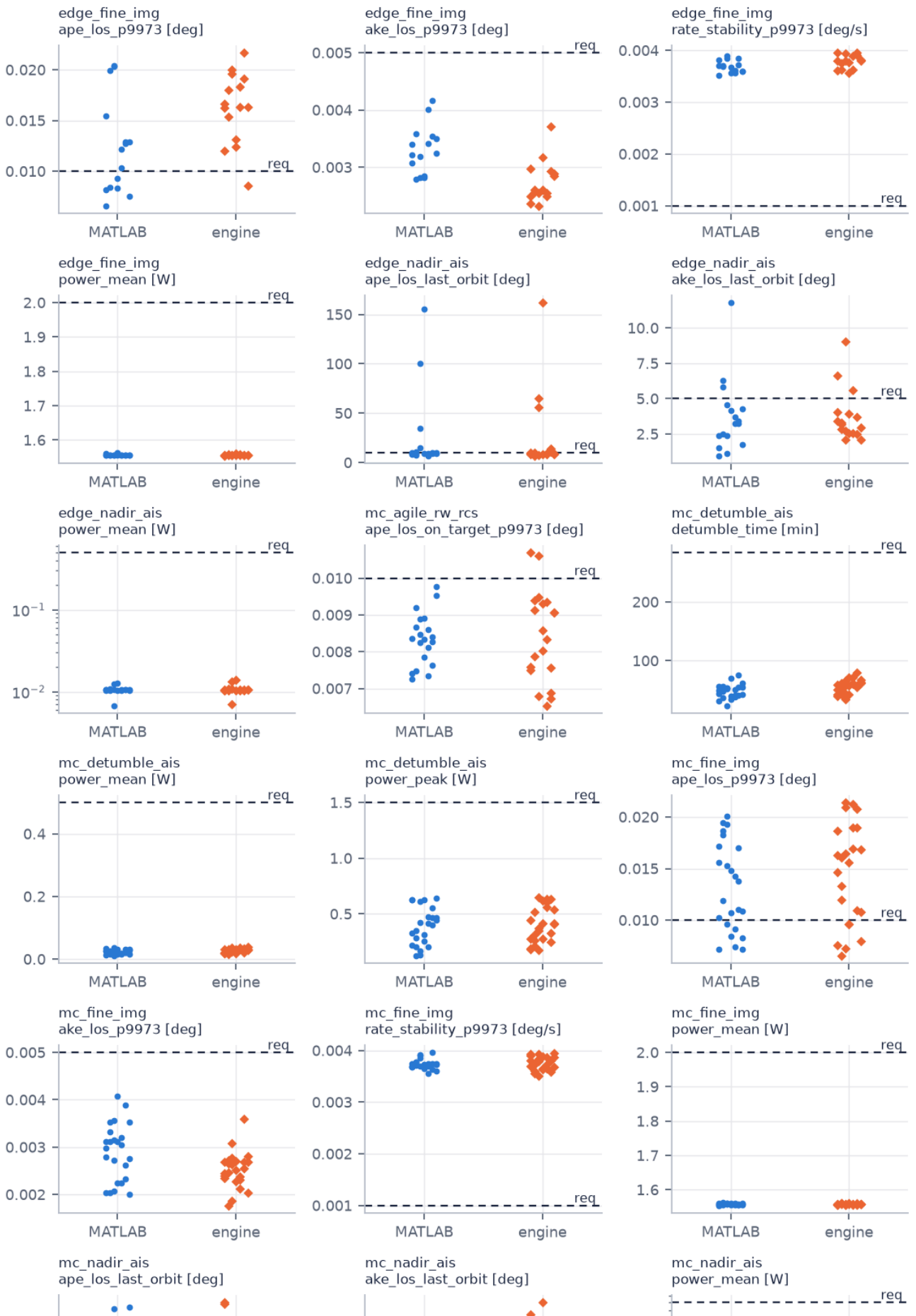
6. Monte Carlo and edge campaigns vs the MATLAB twin

Every campaign of `campaigns/*.toml` flown on the Rust engine with the draws of `asils.campaign.draw` (same dispersions, bounds, run count, per-run seeds), next to the MATLAB twin's campaign. Monte Carlo realisations differ (different random streams) so distributions are compared; edge runs are compared run by run. The engine's flight software keeps the nominal inertia while the plant is dispersed; the MATLAB twin's laws read the dispersed inertia.

campaign	type	metric	req	MATLAB mean $\pm$ std	MATLAB pass	engine mean $\pm$ std	engine pass
edge_fine_img	edge	ape_los_p9973	0.01	0.01236 $\pm$ 0.00475	40 %	0.01625 $\pm$ 0.00351	7 %
edge_fine_img	edge	ake_los_p9973	0.005	0.003307 $\pm$ 0.000418	100 %	0.002739 $\pm$ 0.000362	100 %
edge_fine_img	edge	rate_stability_p9973	0.001	0.003689 $\pm$ 0.000116	0 %	0.003788 $\pm$ 0.00013	0 %
edge_fine_img	edge	power_mean	2	1.558 $\pm$ 0.00216	100 %	1.557 $\pm$ 0.00112	100 %
edge_nadir_ais	edge	ape_los_last_orbit	10	24.91 $\pm$ 40.4	59 %	24.19 $\pm$ 39.4	65 %
edge_nadir_ais	edge	ake_los_last_orbit	5	3.712 $\pm$ 2.58	82 %	3.713 $\pm$ 1.82	82 %
edge_nadir_ais	edge	power_mean	0.5	0.01057 $\pm$ 0.00121	100 %	0.01069 $\pm$ 0.00144	100 %
mc_agile_rw_rcs	montecarlo	ape_los_on_target_p9973	0.01	0.008337 $\pm$ 0.000712	100 %	0.008427 $\pm$ 0.00124	90 %
mc_detumble_ais	montecarlo	detumble_time	284	48.5 $\pm$ 12	100 %	56.29 $\pm$ 11.4	100 %
mc_detumble_ais	montecarlo	power_mean	0.5	0.02267 $\pm$ 0.00701	100 %	0.02764 $\pm$ 0.00684	100 %
mc_detumble_ais	montecarlo	power_peak	1.5	0.3902 $\pm$ 0.169	100 %	0.402 $\pm$ 0.16	100 %
mc_fine_img	montecarlo	ape_los_p9973	0.01	0.01315 $\pm$ 0.00434	29 %	0.01456 $\pm$ 0.00486	25 %
mc_fine_img	montecarlo	ake_los_p9973	0.005	0.002894 $\pm$ 0.000605	100 %	0.002523 $\pm$ 0.000383	100 %
mc_fine_img	montecarlo	rate_stability_p9973	0.001	0.003738 $\pm$ 9.57e-05	0 %	0.003755 $\pm$ 0.00013	0 %
mc_fine_img	montecarlo	power_mean	2	1.558 $\pm$ 0.0021	100 %	1.558 $\pm$ 0.00149	100 %
mc_nadir_ais	montecarlo	ape_los_last_orbit	10	89.69 $\pm$ 38.5	0 %	92.98 $\pm$ 42.3	0 %
mc_nadir_ais	montecarlo	ake_los_last_orbit	5	3.3 $\pm$ 1.73	75 %	2.829 $\pm$ 1.69	88 %
mc_nadir_ais	montecarlo	power_mean	0.5	0.01145 $\pm$ 0.0025	100 %	0.01004 $\pm$ 0.0029	100 %
mc_slew_cmgy	montecarlo	settle_time_after_slew	20	13.14 $\pm$ 0.46	100 %	13.03 $\pm$ 0.785	100 %
mc_slew_cmgy	montecarlo	ape_los_on_target_p9973	0.01	0.003931 $\pm$ 0.000635	100 %	0.003761 $\pm$ 0.000349	100 %

campaign	type	metric	req	MATLAB mean ± std	MATLAB pass	engine mean ± std	engine pass
mc_slew_img	montecarlo	settle_time_after_slew	20	13.03 ± 0.959	100 %	13.84 ± 4.76	98 %
mc_slew_img	montecarlo	ape_los_on_target_p9973	0.01	0.00815 ± 0.00115	92 %	0.007886 ± 0.001	95 %

• MATLAB twin ♦ Rust engine



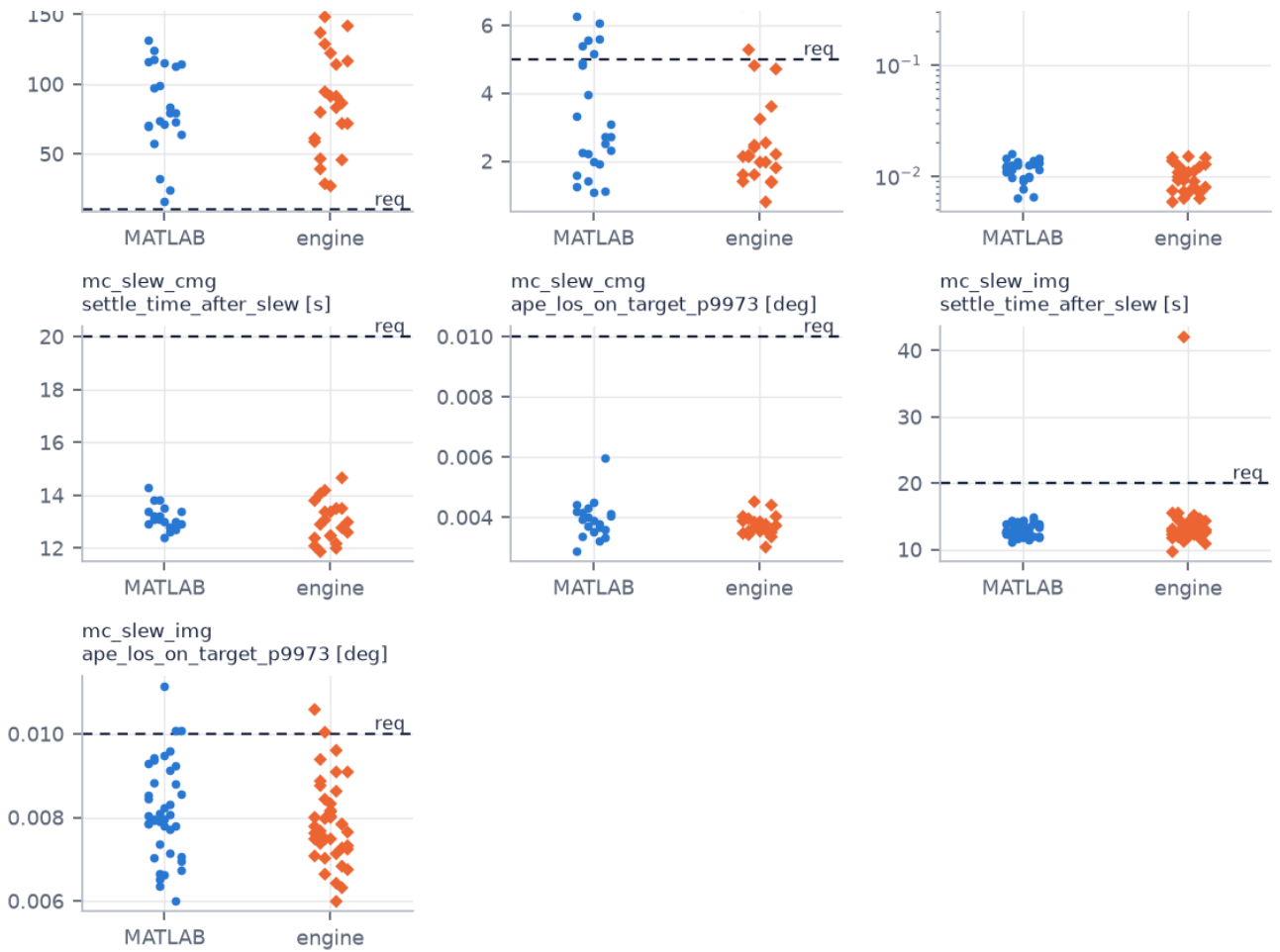


Figure 3. Every run of every campaign, requirement metrics: MATLAB twin (circles) and Rust engine (diamonds), dashed line = requirement.

7. Engine vs MATLAB twin parity

Single runs, engine vs MATLAB twin, same scenario, case and product: **94 of 107** requirement verdicts agree. The truth environment (orbit, Sun, Moon, shadow, density, field) of the engine's POP port is identical to the twin's on every filed sample of every scenario; what differs is the random stream (unit errors, initial tumble), so single-run metrics scatter inside the distributions of section 6. The differences traced to their cause are listed in results/ENGINE_PARITY.md.

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
agile_slew_cmgy	6001	0	0	0	
agile_slew_fmry_rcs	6001	0	0	0	
agile_slew_img	6001	0	0	0	
agile_slew_rw_rcs	6001	0	0	0	
agile_slew_vscmg	6001	0	0	0	
detumble_ais	17221	0	0	0	
detumble_ais_bangbang	17221	0	0	0	
detumble_ais_mag	17221	0	0	0	
detumble_img	17221	0	0	0	
fault_coil_ais	17221	0	0	0	
fault_gimbal_cmgy	57401	0	0	0	
fault_gyro_ais	17221	0	0	0	
fault_st_img	57401	0	0	0	
fault_wheel_img	57401	0	0	0	
fine_hold_cmgy	57401	0	0	0	
fine_hold_fmry	57401	0	0	0	
fine_hold_fmry_rcs	57401	0	0	0	
fine_hold_img	57401	0	0	0	
fine_hold_img_lqr	57401	0	0	0	
fine_hold_img_smc	57401	0	0	0	
fine_hold_rw_rcs	57401	0	0	0	
fine_hold_vscmg	57401	0	0	0	
mission_ais	28701	0	0	0	
mission_cmgy	17221	0	0	0	
mission_fmry	17221	0	0	0	
mission_fmry_rcs	17221	0	0	0	
mission_img	17221	0	0	0	
mission_rw_rcs	17221	0	0	0	
mission_vscmg	17221	0	0	0	
nadir_hold_ais	17221	0	0	0	

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
nadir_hold_ais_css	17221	0	0	0	
slew_cmgy	6001	0	0	0	
slew_fmry	6001	0	0	0	
slew_fmry_rcs	6001	0	0	0	
slew_img	6001	0	0	0	
slew_img_lqr	6001	0	0	0	
slew_img_smc	6001	0	0	0	
slew_rw_rcs	6001	0	0	0	
slew_vscmg	6001	0	0	0	
sun_spin_ais	17221	0	0	0	
agile_slew_cmgy	55	0.14	405x		
agile_slew_fmry_rcs	51	0.14	359x		
agile_slew_img	48	0.12	384x		
agile_slew_rw_rcs	55	0.18	300x		
agile_slew_vscmg	61	0.15	402x		
detumble_ais	496	0.33	1512x		
detumble_ais_bangbang	498	0.33	1524x		
detumble_ais_mag	526	0.32	1660x		
detumble_img	972	1.91	509x		
fault_coil_ais	533	0.32	1650x		
fault_gimbal_cmgy	465	1.14	406x		
fault_gyro_ais	579	0.38	1535x		
fault_st_img	389	0.80	486x		
fault_wheel_img	414	1.16	357x		
fine_hold_cmgy	468	1.01	462x		
fine_hold_fmry	659	1.00	658x		
fine_hold_fmry_rcs	523	0.99	531x		
fine_hold_img	390	0.98	398x		
fine_hold_img_lqr	388	1.09	355x		
fine_hold_img_smc	388	0.99	390x		
fine_hold_rw_rcs	394	1.00	394x		
fine_hold_vscmg	476	1.08	439x		
mission_ais	906	0.55	1647x		
mission_cmgy	1421	2.68	530x		
mission_fmry	1622	2.42	670x		
mission_fmry_rcs	1619	2.74	590x		
mission_img	1235	2.72	455x		
mission_rw_rcs	1263	2.65	476x		

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
mission_vscmg	1461	2.97	492x		
nadir_hold_ais	569	0.36	1573x		
nadir_hold_ais_css	550	0.40	1374x		
slew_cmg	54	0.17	314x		
slew_fmr	79	0.15	509x		
slew_fmr_rcs	56	0.15	383x		
slew_img	46	0.15	298x		
slew_img_lqr	46	0.15	311x		
slew_img_smc	45	0.15	307x		
slew_rw_rcs	46	0.15	318x		
slew_vscmg	56	0.15	381x		
sun_spin_ais	501	0.32	1561x		

Verdicts that differ (all traced; see ENGINE_PARITY.md):

scenario	metric	MATLAB	engine	req	MATLAB	engine
detumble_ais_bangbang	detumble_time	—	32.21	284	FAIL	pass
fine_hold_fmr	power_mean	4.348	0.08406	2	FAIL	pass
fine_hold_fmr_rcs	power_mean	4.367	0.08406	2	FAIL	pass
fine_hold_img	ape_los_p9973	0.009166	0.01049	0.01	pass	FAIL
fine_hold_rw_rcs	ape_los_p9973	0.008375	0.03388	0.01	pass	FAIL
mission_img	ape_los_last_half_orbit_p9973	0.007627	0.01066	0.01	pass	FAIL
mission_vscmg	ape_los_last_half_orbit_p9973	0.01111	0.009595	0.01	FAIL	pass
nadir_hold_ais	ape_los_last_orbit	7.248	12.64	10	pass	FAIL
nadir_hold_ais	ake_los_last_orbit	3.864	5.491	5	pass	FAIL
slew_img_smc	settle_time_after_slew	3.2	—	20	pass	FAIL
sun_spin_ais	sun_spin_entry	40.44	142.4	95	pass	FAIL
sun_spin_ais	sun_angle_last_orbit	6.503	93.84	20	pass	FAIL
sun_spin_ais	spin_rate_error_last_orbit	0.04568	1.116	0.5	pass	FAIL

8. Soft OILS

Every scenario flown twice on the engine: SILS (C flight software in-process) and soft OILS (the flight software compiled for the OBC, arm-none-eabi-gcc -mcpu=cortex-m4 -mfloat-abi=hard, running as firmware on QEMU mps2-an386 behind adcs-link/1). In soft OILS each command reaches the actuators only after the OBC's sensor reads, its execution and its CAN frames; the execution is the step's exact instruction count (QEMU plugin) $\times$ CPI 1.25 / 168 MHz. **104 of 107** requirement verdicts agree between SILS and soft OILS over 40 scenarios; runs are deterministic.

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
agile_slew_cmgy	100	3.5e+05 / 8.95e+05	6.66	4.2 / 8.25	
agile_slew_fmr_rcs	100	3.01e+05 / 8.45e+05	6.29	3.31 / 7.36	
agile_slew_img	100	2.95e+05 / 8.43e+05	6.27	3.14 / 7.22	
agile_slew_rw_rcs	100	3.04e+05 / 8.52e+05	6.34	3.34 / 7.41	
agile_slew_vscmg	100	3.68e+05 / 9.16e+05	6.81	4.33 / 8.41	
detumble_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_ais_bangbang	200	1.17e+05 / 4.64e+05	3.45	1.42 / 4	
detumble_ais_mag	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_img	100	8.12e+04 / 4.89e+05	3.64	1.55 / 4.58	
fault_coil_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
fault_gimbal_cmgy	100	3.62e+05 / 8.97e+05	6.68	4.29 / 8.27	
fault_gyro_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
fault_st_img	100	2.54e+05 / 8.42e+05	6.27	2.84 / 7.21	
fault_wheel_img	100	2.9e+05 / 8.44e+05	6.28	3.1 / 7.23	
fine_hold_cmgy	100	3.62e+05 / 8.97e+05	6.68	4.29 / 8.27	
fine_hold_fmr	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_fmr_rcs	100	3.13e+05 / 8.48e+05	6.31	3.4 / 7.38	
fine_hold_img	100	3.1e+05 / 8.46e+05	6.29	3.25 / 7.23	
fine_hold_img_lqr	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_img_smc	100	3.13e+05 / 8.48e+05	6.31	3.27 / 7.26	
fine_hold_rw_rcs	100	3.15e+05 / 8.53e+05	6.34	3.42 / 7.42	
fine_hold_vscmg	100	3.83e+05 / 9.19e+05	6.84	4.44 / 8.43	
mission_ais	200	1.86e+05 / 7.58e+05	5.64	1.94 / 6.2	
mission_cmgy	100	3.05e+05 / 8.97e+05	6.67	3.87 / 8.27	
mission_fmr	100	2.61e+05 / 8.46e+05	6.3	2.89 / 7.24	
mission_fmr_rcs	100	2.64e+05 / 8.5e+05	6.32	3.04 / 7.4	
mission_img	100	2.68e+05 / 8.45e+05	6.29	2.94 / 7.23	
mission_rw_rcs	100	2.74e+05 / 8.55e+05	6.36	3.12 / 7.43	
mission_vscmg	100	3.23e+05 / 9.18e+05	6.83	3.99 / 8.43	

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
nadir_hold_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
nadir_hold_ais_css	200	1.94e+05 / 6.38e+05	4.74	1.99 / 5.3	
slew_cmng	100	3.69e+05 / 8.94e+05	6.65	4.34 / 8.24	
slew_fmr	100	3.17e+05 / 8.42e+05	6.27	3.3 / 7.21	
slew_fmr_rcs	100	3.19e+05 / 8.45e+05	6.29	3.45 / 7.36	
slew_img	100	3.17e+05 / 8.42e+05	6.26	3.3 / 7.21	
slew_img_lqr	100	3.17e+05 / 8.42e+05	6.26	3.3 / 7.21	
slew_img_smc	100	3.19e+05 / 8.45e+05	6.29	3.32 / 7.23	
slew_rw_rcs	100	3.21e+05 / 8.47e+05	6.3	3.46 / 7.38	
slew_vscmg	100	3.89e+05 / 9.15e+05	6.81	4.49 / 8.4	
sun_spin_ais	200	1.2e+05 / 4.65e+05	3.46	1.45 / 4.01	

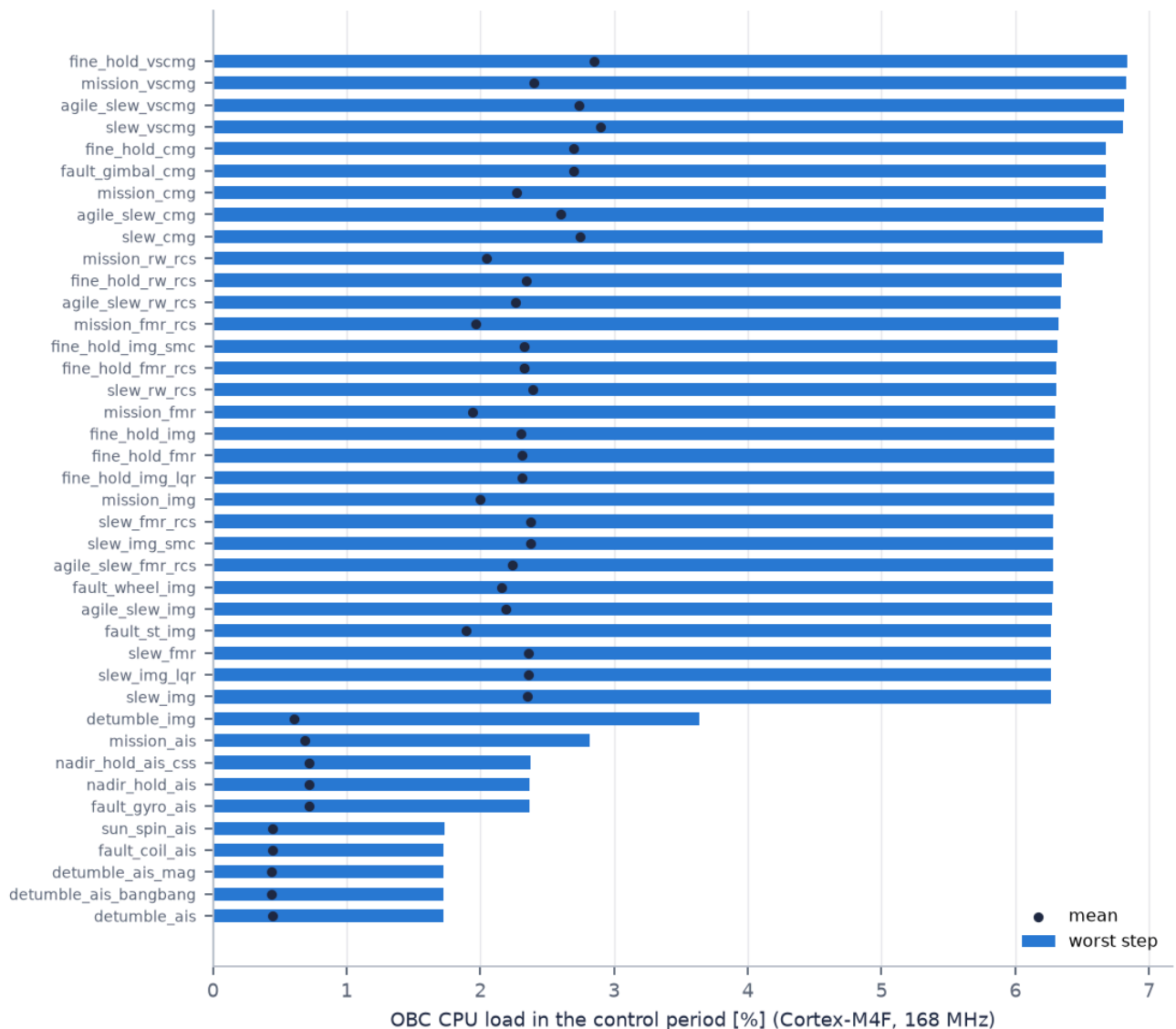


Figure 4. OBC load per scenario: worst step (bar) and mean (dot).

Verdicts that change between SILS and soft OILS (the command latency's effect):

scenario	metric	req	SILS	soft OILS	SILS	soft OILS
agile_slew_vscmg	ape_los_on_target_p9973	0.01	0.01008	0.009971	FAIL	pass
detumble_ais_bangbang	detumble_time	284	32.21	—	pass	FAIL
sun_spin_ais	sun_spin_entry	95	142.4	61.91	FAIL	pass

9. Virtual OBC bit-identity

Zero-latency lockstep (no timing model): the flight software as a host process and as QEMU Cortex-M4F firmware, C and Rust, reproduces the in-process trajectory bit for bit: **35 of 35** runs bit-identical.

scenario	reference	virtual OBC	result	wall [s]
detumble_ais	c (in-process)	obc-posix	bit-identical	0.4
detumble_ais	rust (in-process)	obc-posix-rs	bit-identical	0.3
detumble_ais	c (in-process)	qemu	bit-identical	21.6
detumble_ais	rust (in-process)	qemu-rs	bit-identical	22.0
detumble_ais	c (in-process)	rust	bit-identical	0.1
mission_ais	c (in-process)	obc-posix	bit-identical	0.3
mission_ais	rust (in-process)	obc-posix-rs	bit-identical	0.3
mission_ais	c (in-process)	qemu	bit-identical	23.3
mission_ais	rust (in-process)	qemu-rs	bit-identical	22.3
mission_ais	c (in-process)	rust	bit-identical	0.1
fine_hold_img	c (in-process)	obc-posix	bit-identical	0.7
fine_hold_img	rust (in-process)	obc-posix-rs	bit-identical	0.7
fine_hold_img	c (in-process)	qemu	bit-identical	74.1
fine_hold_img	rust (in-process)	qemu-rs	bit-identical	76.4
fine_hold_img	c (in-process)	rust	bit-identical	0.4
slew_img	c (in-process)	obc-posix	bit-identical	0.8
slew_img	rust (in-process)	obc-posix-rs	bit-identical	0.7
slew_img	c (in-process)	qemu	bit-identical	71.6
slew_img	rust (in-process)	qemu-rs	bit-identical	81.8
slew_img	c (in-process)	rust	bit-identical	0.4
fine_hold_cmng	c (in-process)	obc-posix	bit-identical	0.8
fine_hold_cmng	rust (in-process)	obc-posix-rs	bit-identical	0.7
fine_hold_cmng	c (in-process)	qemu	bit-identical	83.1
fine_hold_cmng	rust (in-process)	qemu-rs	bit-identical	81.7
fine_hold_cmng	c (in-process)	rust	bit-identical	0.4
fine_hold_fmr_rcs	c (in-process)	obc-posix	bit-identical	0.7
fine_hold_fmr_rcs	rust (in-process)	obc-posix-rs	bit-identical	0.9
fine_hold_fmr_rcs	c (in-process)	qemu	bit-identical	77.0
fine_hold_fmr_rcs	rust (in-process)	qemu-rs	bit-identical	74.8

scenario	reference	virtual OBC	result	wall [s]
fine_hold_fmr_rcs	c (in-process)	rust	bit-identical	0.4
sun_spin_ais	c (in-process)	obc-posix	bit-identical	0.3
sun_spin_ais	rust (in-process)	obc-posix-rs	bit-identical	0.3
sun_spin_ais	c (in-process)	qemu	bit-identical	20.3
sun_spin_ais	rust (in-process)	qemu-rs	bit-identical	24.1
sun_spin_ais	c (in-process)	rust	bit-identical	0.1

10. Open items

- **ais_3u**: the selected `mtq_fmr` leaves budget: mass_kg 0.855 > 0.35; budget: volume_L 0.477 > 0.3. Either the case relaxes these (several are marked UNCONFIRMED in the case file) or the next design lever is needed.
- **ais_img_3u**: the selected `mtq_fmr` leaves nadir_pointing: ape_los_p9973 (performance), rate_stability_p9973 (performance); budget: mass_kg 1.56 > 1. Either the case relaxes these (several are marked UNCONFIRMED in the case file) or the next design lever is needed.
- Soft OILS finding: bang-bang B-dot (detumble_ais_bangbang) detumbles in 32 min in SILS but, with the OBC's 2-8 ms command latency, settles into a 0.7-1.0 deg/s limit cycle and never holds below 0.5 deg/s. The proportional B-dot laws are unaffected. Keep the proportional law as the flight default, or add a dead band / latency compensation to the bang-bang law before OILS.
- Soft OILS knife edges (verdict flips, not trends): agile_slew_vscmg APE 0.01008 vs 0.00997 deg against 0.01 deg; sun_spin_ais entry time (the known sign-flip lock-up, which the latency happens to avoid in this seed).
- Fine hold (imaging 3U): rate stability p99.73 is $\sim 3.7e-3$ deg/s against $1e-3$ deg/s in every configuration and every campaign (MATLAB and engine alike): a requirement/controller-bandwidth conflict to resolve before OILS.
- Fluid-loop field power: the 2 W electromagnet pump dominates the power of our FMR families; the permanent-magnet yoke removes it (the loop's power lever). It needs a measured prototype (IDMAS v2 §03C).
- Sun-spin sign flip: a spin reversal instead of a hemisphere manoeuvre locks up in 3 of 24 seeds (identical in MATLAB, C and Rust).
- Soft OILS models the OBC's CPU and buses; real OILS still needs the board target (fsw/targets/<board>/main.c: UART/Ethernet, clock, linker script, modelled on fsw/targets/qemu-mps2) and --realtime on adcs-link/1.